Technical review

Item #2 · WhatsApp Chat with Ana Ruiz.txt · Run ID: 20261113T232759Z-440d07

Path	R:\evidence\chat\whatsapp-android-en\WhatsApp Chat with Ana Ruiz.txt
Size (bytes)	773
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.txt
Detected type (signature)	unknown
Status	Needs attention
Case	RV-SAMPLE-001
Evidence ID	E0010
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`0642f460332ab8b5887be935dfefbef8214786c2247ec98d4c5eed832cbdb71d`
md5	`e59d7e7df7723088be26d481a367eabe`

RSMF conversion

This export was converted into 3 RSMF file(s) holding 10 event(s) from 1 conversation(s); every file passed the RSMF 2.0 schema, the stricter checks and the read-back test.

Export type	whatsapp
Participants	3
Events	10
chat_file	WhatsApp Chat with Ana Ruiz.txt
date_order	dmy
date_order_basis	line 11: first component 13 > 12
lines_read	13
timezone	Europe/Paris
variant	android
File	`001-WhatsApp-Chat-with-Ana-Ruiz-txt-000002/Ana-Ruiz-aa3173/2026-03-02.rsmf`
Events	7 · 1 attachments · passed
SHA-256	`4527fa7e7ca432ccc5d7599e6fb7678f5fca938bfc7c145fdb5030ca2df9ece3`
File	`001-WhatsApp-Chat-with-Ana-Ruiz-txt-000002/Ana-Ruiz-aa3173/2026-03-13.rsmf`
Events	2 · 0 attachments · passed
SHA-256	`21ce0b8c75901c6266aca3a14e68ef8167572a92610767c3f88b92ce3f52698c`
File	`001-WhatsApp-Chat-with-Ana-Ruiz-txt-000002/Ana-Ruiz-aa3173/2026-03-14.rsmf`
Events	1 · 0 attachments · passed
SHA-256	`0854504d5f9d9facbc4f867759054707fc485240e4d9343b31aceec363d6177a`

Every message is listed in rsmf/trace.csv with the file and line it came from and the RSMF file and event it went to; each .rsmf file's SHA-256 is in the audit log (rsmf_written).

We uncover the hidden. We reconstruct what happened. We protect what matters.

Findings

[Informational] Date order inferred from the export (DATE_ORDER_INFERRED)

What it shows	Numeric dates were read as dmy: line 11 ("13/03/2026, 18:45") can only be read that way.
What it does not prove	It does not show that every line was written with the same phone settings; an export that mixes day-first and month-first dates stops with an error instead of being converted.

[Low] Media not included in the export (MEDIA_OMITTED_IN_EXPORT)

What it shows	1 message(s) say the media was omitted when the chat was exported ("without media").
What it does not prove	It does not show what the media contained, or that it was deleted from the phone.

[Medium] Referenced file not found (ATTACHMENT_REFERENCED_NOT_FOUND)

What it shows	1 attachment(s) are named in the export but the file is not in it: IMG-20260302-WA0002.jpg.
What it does not prove	It does not show that the file was removed on purpose; exports can be incomplete.

[Informational] Participants known only by phone number (PARTICIPANT_IDENTIFIED_BY_NUMBER_ONLY)

What it shows	1 participant(s) appear only as a number (not in the exporting phone's contacts): +33 6 00 00 00 01.
What it does not prove	A number does not identify a person; establish who used it from other evidence.

[Informational] Local times converted with the examiner's time zone (TIMEZONE_APPLIED)

What it shows	The export's times carry no UTC offset; they were read as Europe/Paris and stored in UTC. The original text of each time is kept in the event's custom field source_time.
What it does not prove	It does not show that the phone was set to Europe/Paris; a wrong zone shifts every time by the same amount.

[Informational] Deleted messages are not exported (DELETED_MESSAGES_NOT_IN_EXPORT)

What it shows	WhatsApp exports leave out messages deleted before the export; the RSMF can only contain what the export contains.
What it does not prove	It does not show whether any messages were deleted.

Method

Each file was opened read-only and hashed in a single pass with the selected algorithms. Chat exports were read into one model (participants, conversations, messages, attachments), local times converted to UTC with the examiner's time zone, and written as RSMF 2.0 files, one per conversation per day, then checked against the RSMF 2.0 schema and read back. Every action was written to a hash-chained audit log.

Audit trail

Audit entries	#9 – #12
Log folder	`R:\case\runs\rv-rsmf\20261113T232759Z-440d07\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- The RSMF files reproduce the export as delivered, not the phone or account: messages deleted before the export, unexported media and edit history are not in them.
- rv-rsmf converts; it does not authenticate. A chat export is a text or JSON file that can be edited before it reaches the examiner; corroborate it with a forensic extraction or the other party's copy.
- Files are checked against the published RSMF 2.0 schema and read back; how a given Relativity version displays them was not tested by this run.
- Reactions are mapped with a fixed table of Relativity emoji shortcodes (checked 2026-10-27); other reactions are kept as text.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.